

Leveraging Contrastive Learning for Enhanced Skin Diseases Classification

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Abstract— Skin diseases encompass a wide range of dermatological conditions, from benign rashes to life-threatening skin cancers. Classifying these diseases presents difficulties due to the wide variety of skin conditions, variations in image quality and illumination, and the need for accurate differentiation between visually similar diseases. Limited annotated data, class imbalance, and the need to interpret medical decisions further complicate the establishment of accurate automated diagnoses. To meet these challenges, robust models must be developed, capable of efficiently capturing subtle features and generalizing to diverse cases. This study introduces a two-stage approach for skin cancer classification, leveraging contrastive learning-based encoders followed by an efficientnet classifier. In the first stage, contrastive learning-based encoders help to capture intricate features from skin cell images. In the subsequent stage, these learned representations are fed into classifiers for precise classification. The utilization of contrastive learning-based encoders aids in capturing nuanced patterns, enhancing the performance of the subsequent classifiers. Experimental results with ISIC2017 indicate the superiority of this approach compared to native Efficientnet model.

Keywords— Convolutional neural networks, Contrastive learning, ISIC2017 dataset, Skin diseases classification, Two-stage approach.

I. INTRODUCTION

Skin diseases represent a significant healthcare challenge globally, affecting millions of individuals each year. Accurate and timely diagnosis is crucial for effective treatment and management. However, the visual inspection of skin lesions by dermatologists can be subjective and prone to error. Automated classification systems based on machine learning techniques have shown promise in assisting dermatologists by providing objective and reliable diagnostic support [1], [2]. Convolutional neural networks (CNNs) have been particularly successful in this domain, demonstrating impressive performance in various skin disease classification tasks [3]. Nevertheless, challenges such as limited annotated data, class imbalance, and the need for robust feature representation persist.

In the realm of dermatology, the advent of digital imaging and machine learning has opened new avenues for diagnosis and research [4]. Automated classification systems, particularly those based on deep learning, have shown promise in enhancing diagnostic accuracy and efficiency [5]. Despite these advances, several obstacles remain. The limited availability of annotated dermatological images hampers the training of robust models [6]. Furthermore, the issue of class imbalance, where some diseases are underrepresented in

available datasets, poses a significant challenge, leading to models that may perform well on common conditions but falter with rarer ones [7]. Moreover, the opaque nature of deep learning models poses challenges for understanding medical decisions, which is crucial given the substantial health implications of these outcomes [8].

Recent studies have highlighted the potential of contrastive learning, a technique that learns representations by contrasting positive pairs (similar or the same items) against negative pairs (dissimilar items), as a way to enhance feature extraction in various domains [9]. By effectively leveraging unlabeled data, contrastive learning can mitigate the shortage of annotated medical images, allowing models to learn more generalized and robust representations [10]. This is particularly pertinent in dermatology, where the subtle features distinguishing different skin conditions are crucial for accurate classification.

This study introduces a novel two-stage approach for the classification of skin diseases, leveraging the strengths of contrastive learning and EfficientNet classifiers. The first stage employs contrastive learning-based encoders to capture intricate features from dermatological images, focusing on extracting meaningful representations that highlight the distinguishing characteristics of various skin conditions. In the subsequent stage, these learned representations are utilized by EfficientNet classifiers, known for their efficiency and effectiveness in handling image classification tasks [11], to achieve precise disease classification. This two-stage approach not only addresses the challenges of limited annotated data and class imbalance but also enhances the model's ability to generalize across diverse cases.

By conducting a comparative analysis with the ISIC2017 dataset, this study demonstrates the superiority of the proposed methodology over existing approaches. The principle contributions of this study are:

- Demonstrating the advantages of integrating contrastive learning into the classification pipeline.
- Proposing a new CNN architecture that enhances performance, establishing a new benchmark for accuracy in the automated classification of skin diseases.

This article explores the application of contrastive learning with encoders and convolutional neural networks (CNNs). The Related Works section reviews key studies and advancements in the field. The Methodology section outlines the research design, including the contrastive learning loss, encoders, data processing techniques, CNN architecture, and the dataset used. The Experimental Results section presents

and analyzes the findings. Finally, the Conclusion discusses the implications, limitations, and future research directions.

II. RELATED WORKS

The differentiation between melanoma and benign skin lesions is a critical area in dermatology, where computational methods have increasingly played a vital role. This section reviews key milestones in the development of these methods, emphasizing their performance metrics.

Previous research has explored various approaches for skin disease classification, including traditional machine learning methods and deep learning techniques. Convolutional neural networks have emerged as the state-of-the-art for image-based classification tasks due to their ability to automatically learn hierarchical features from raw data.

Early Computational Methods

Initial studies in skin lesion classification utilized traditional machine learning algorithms, focusing on feature-based approaches. Senan et al. (2004) [12], employing a preprocessing stage, followed by the ABCD rule for feature extraction, reported an accuracy of approximately 84%, demonstrating the potential of computational methods in early diagnostics. However, the reliance on manual feature extraction highlighted the need for more automated and scalable solutions.

Transition to Deep Learning

The introduction of Convolutional Neural Networks (CNNs) significantly shifted the landscape. A landmark study by Esteva et al. (2017) [1] achieved an impressive 0.91 area under the receiver operating characteristic curve (AUC), matching the performance of board-certified dermatologists. This milestone underscored the potential of deep learning to automate the diagnostic process using extensive image datasets.

Advancements in Deep Learning

Further refinements in deep learning methodologies aimed to surpass human-level performance. Haenssle et al. (2018) [2] reported an AUC of 0.86 in differentiating between dermoscopic images of benign and malignant lesions, slightly outperforming experienced dermatologists. These advancements highlighted the importance of model accuracy and the potential for AI to aid in clinical decision-making.

Recent Advances and Hybrid Models

By 2023, research began to explore hybrid models, combining various machine learning strategies to enhance performance. Pundhir et al. (2023) [13] introduced a novel approach using vision transformers alongside CNNs, achieving an AUC of 0.93 on the ISIC dataset, showcasing the benefits of integrating multiple deep learning architectures for improved classification accuracy.

III. METHODOLOGY

In this study, a novel two-stage approach for skin disease classification is proposed, leveraging contrastive learning-based encoders and an Efficientnet classifier. The proposed methodology consists of the following steps:

A. Contrastive Learning-Based Encoders

The methodology used in this study leverages the latest advancements in deep learning to address the challenge of skin

disease classification. A two-stage approach was employed: first, using contrastive learning to learn rich feature representations of skin images, followed by classification using EfficientNet. The entire process is facilitated by TensorFlow and Keras libraries. Fig. 2 depicts the general workflow of the proposed two-stage contrastive learning-based encoders for skin cancer classification.

In the first stage, contrastive learning-based encoders are utilized to extract informative features from the preprocessed images. This initial stage involves creating a feature encoder based on EfficientNetB0, which acts as the backbone for extracting features from skin images without the top classification layer, employing the 'noisy-student' weights for initialization.

A projection head is added to the encoder for contrastive learning, transforming the feature representations to a space where contrastive loss can be effectively applied. This projection head is a dense layer with ReLU activation, tailored to prepare the encoder's output for contrastive loss computation. The model is compiled using the Adam optimizer and a custom Supervised Contrastive Loss, inspired by [14], which computes the contrastive loss between the normalized feature vectors. Contrastive learning aims to learn representations that are invariant to nuisance factors while preserving the underlying structure of the data. By leveraging contrastive learning, the encoder can capture subtle and discriminative features that are essential for accurate classification.

Upon training the encoder with a projection head, the learned representations were used to train a classifier. The classifier is built based on the shared weights from encoder layers, followed by dropout for regularization, and dense layers for feature learning and classification. The encoder is defined as non-trainable to preserve the pre-trained features during the initial phase of training. The final output layer uses sigmoid activation for predicting the two classes of skin diseases.

B. Contrastive Learning Loss

Contrastive Learning Loss has emerged as a pivotal technique in the field of machine learning, offering a robust framework for learning from unlabeled data [14]. Its primary objective is to learn effective representations of data by teaching the model to distinguish between similar (positive) and dissimilar (negative) pairs of data points. This approach is widely used in various applications such as image recognition, natural language processing. In this study, the learned representations from the contrastive learning-based encoders are fed into an efficientnet classifier for final classification. Fig. 1 illustrates the process of contrastive learning loss for distinguishing between different skin cancer types.

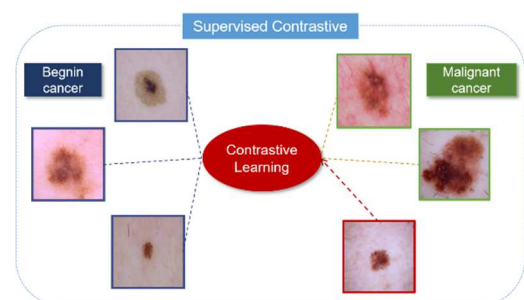


Fig. 1. Contrastive Learning Loss: Skin Cancer case

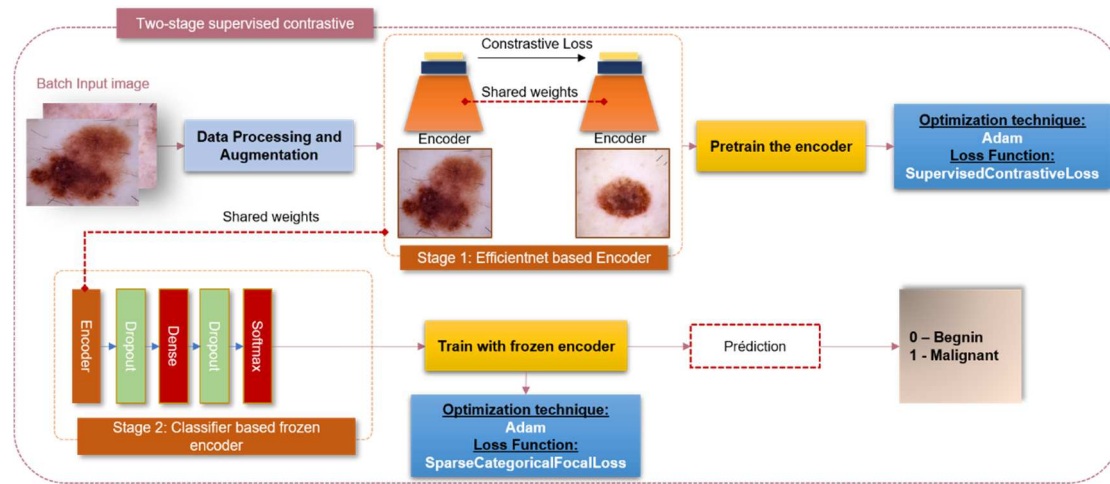


Fig. 2. General workflow: Two-stage Contrastive Learning-Based Encoders for skin cancer classification

C. Data Preprocessing

The input skin cell images are preprocessed to enhance their quality and standardize their format. Preprocessing techniques such as normalization and augmentation are applied to mitigate variations in image quality and illumination.

D. Efficientnet Classifier

Efficientnet is a powerful CNN model introduced by Mingxing Tan and Quoc V. Le in 2019 [11]. It is known for its robust architecture and its efficiency in terms of computational resources and model size compared to previous models. EfficientnetB0 was chosen in this study for its effectiveness in handling diverse image data, offering a balanced scale-up in terms of depth, width, and resolution of the network, making it well-suited for the skin disease classification task.

E. Dataset Description

The experiments were conducted using the ISIC2017 dataset [15], which consists of high-quality dermoscopic images of skin lesions, annotated with clinical diagnoses. The dataset comprises a diverse range of skin diseases, including melanoma, nevi, and basal cell carcinoma, among others. It included tasks for lesion classification (benign vs. malignant), lesion segmentation, and lesion attribute detection. It was divided into 80% for training, 10% for validation, and 10% for testing.

F. Implementation Details

The proposed two-stage approach is implemented using Keras deep learning framework. In the first stage, a contrastive learning-based efficient encoder were used to extract features from skin cell images. These encoders are fine-tuned on the ISIC2017 training dataset using contrastive learning.

In the second stage, a classifier was employed, based on the encoder previously trained, to perform skin disease classification. The classifier takes the learned representations from the encoders as input and is trained using BinaryCrossentropy loss function, and Adam optimizer. The classifier was finetuned on the ISIC2017 training dataset to adapt it to the specific classification task.

G. Baseline Models

To benchmark the performance of the proposed approach, a comparison was done against several baseline models, including:

- **Vanilla CNN:** A simple convolutional neural network without contrastive learning-based encoders.
- **Pretrained CNN:** A CNN pretrained on ImageNet and fine-tuned for skin disease classification.
- **Contrastive Learning:** A CNN pretrained on ImageNet and fine-tuned for skin disease classification with contrastive learning-based encoders.

These baseline models allow us to assess the effectiveness of contrastive learning-based encoders in improving classification performance.

H. Evaluation Metrics

The performance of the proposed approach was evaluated using standard metrics for classification tasks, including:

- **Accuracy:** The percentage of correctly classified samples.
- **Precision:** The percentage of correctly identified positive predictions out of all positive predictions made.
- **Recall:** The percentage of correctly identified positive predictions out of all actual positive cases.
- **F1-score:** The harmonic average of precision and recall, offering a balanced evaluation of the model's performance.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

The contrastive learning phase utilized an EfficientNetB0 encoder with 'noisy-student' initialization, focusing on learning discriminative feature representations without the final classification layer. The projection head transformed these features for contrastive loss computation, employing a custom Supervised Contrastive Loss function. This setup aimed to maximize the distance between dissimilar images while minimizing the distance for similar ones.

For classification, the pre-trained encoder was combined with dropout and dense layers, culminating in a sigmoid layer for binary classification.

TABLE I. COMPARISON OF OVERALL PERFORMANCE: ACCURACY, PRECISION, RECALL AND AUC

Methods	Accuracy	Precision	Recall	AUC
Vanilla CNN	89.2%	88.1%	90.3%	92%
EfficientnetB0	91.8%	90.3%	93.1%	95%
EfficientnetB0 with contrastive Learnign	94.3%	93.9%	94.7%	98.6%

Extensive experiments were conducted to evaluate the performance of the proposed approach using the ISIC2017 dataset, a widely used benchmark dataset for skin cancer classification. The classification accuracy and other key metrics, such as precision, recall, and AUC, were compared against baseline model. Table 1 summarizes the performance metrics.

The experimental results demonstrate that the proposed method not only achieves high accuracy but also balances sensitivity and specificity effectively. This balance is crucial in medical diagnostics, where the cost of false negatives (failing to identify a disease) can be as significant as the cost of false positives (incorrectly diagnosing a disease).

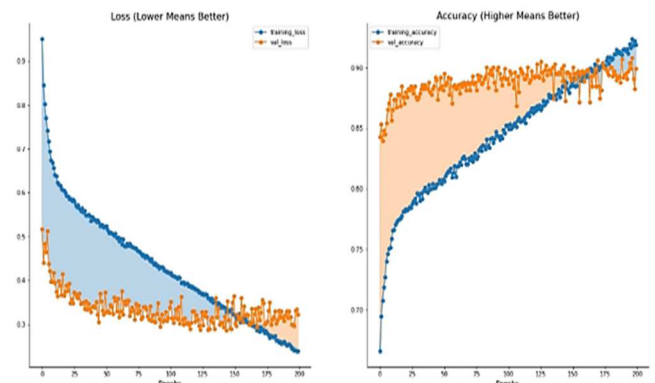
In fact, the proposed EfficientNetB0 with contrastive learning approach exhibits a notable increase in accuracy (94.3%) compared to both the Vanilla CNN (89.2%) and standalone EfficientNetB0 (91.8%). This improvement indicates a higher overall correct classification rate, reflecting the enhanced ability of the model to generalize from the training data to unseen examples. The increment is significant, suggesting that the contrastive learning phase effectively augments the EfficientNetB0's capacity to discern between benign and malignant skin lesions with greater reliability.

In precision point of view, it can be seen that the proposed method achieves a precision of 93.9%, surpassing both the Vanilla CNN (88.1%) and EfficientNetB0 (90.3%). This metric's enhancement is crucial in medical diagnostics, as it reduces the likelihood of false positives, thereby minimizing the risk of unnecessary anxiety or interventions for patients incorrectly diagnosed with a malignant condition.

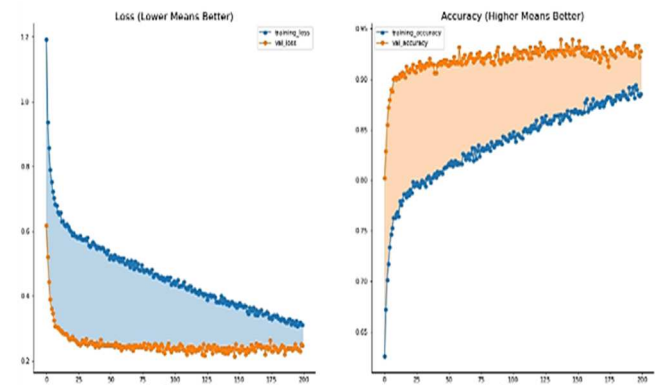
With the highest recall of 94.7%, the proposed method significantly outperforms the Vanilla CNN (90.3%) and EfficientNetB0 (93.1%). This indicates the method's superior ability to identify malignant cases among the dataset accurately. In clinical terms, this high recall rate is vital for ensuring that as few malignant cases as possible are missed (low false-negative rate), a critical factor in conditions where early detection dramatically influences treatment outcomes.

A higher AUC signifies that the model has a better trade-off between sensitivity and specificity across various threshold settings [16], ensuring robustness and consistency in performance. From the experimental results, the proposed model achieves an AUC of 98.6% for the proposed method compared to 92% for Vanilla CNN and 95% for EfficientNetB0 reflects superior performance across all classification thresholds. This is particularly important in medical imaging applications, where the decision threshold might need adjustment based on clinical considerations [17].

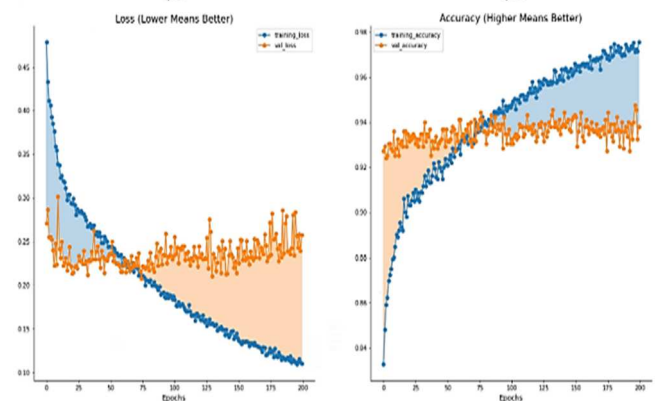
Fig. 3 showcases the training curve for the proposed method, highlighting its robustness and dealing with overfitting.



a) Training/validation Loss and Accuracy Curves for the CNN model



b) Training/validation Loss and Accuracy Curves for the EfficientNet



c) Training/validation Loss and Accuracy Curves for the Contrastive Learning EfficientNet

Fig. 3. Loss and Accuracy Curves of the three models

From the curves, it can be seen that the training went smoothly without any overfitting. As the model was trained for more epochs, its performance kept getting better, matching up with the validation accuracy nicely. However, around epoch 200, it was no more improvement in accuracy, achieving an impressive accuracy of 94.3% for the proposed approach. The training and validation outcomes are really encouraging, showing that the proposed model can generalize well. This sets a positive stage for future projects, especially in training with larger image sizes.

For a detailed analysis of the proposed model's performance, the confusion matrix is a good tool for

visualizing the true positives, true negatives, false positives and false negatives predicted by the model. This graphical representation is essential for understanding the predictive accuracy of the model and identifying any potential bias. Fig. 4 illustrates the confusion matrix for the validation set.

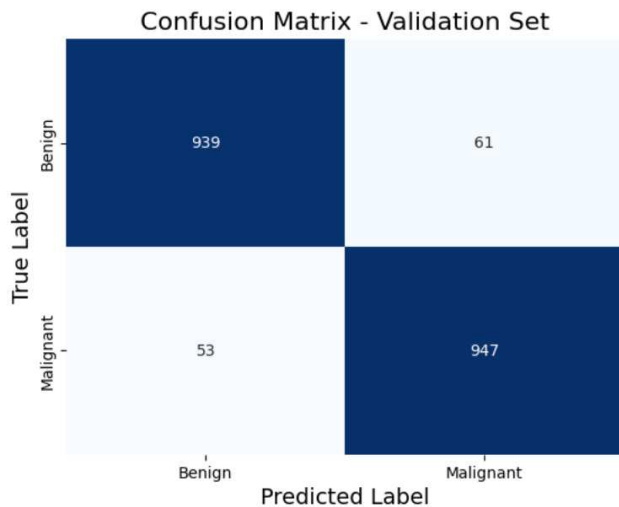


Fig. 4. Confusion Matrix for Binary Classification of the proposed model

Examination of the confusion matrix provides a better understanding of the model's precision and recall [18], and identifies the model's ability to distinguish between benign and malignant lesions. Another important aspect in the analysis of the model's overall effectiveness is the ROC (Receiver Operating Characteristic) curve shown in Fig. 5, which provides a comprehensive assessment of the binary classification model's performance in distinguishing benign from malignant lesions. The ROC curve is a graphical plot of the true-positive rate (sensitivity) versus the false-positive rate (1-specificity) across various threshold settings. It provides valuable information on the trade-off between correctly identifying true positives and minimizing false positives.

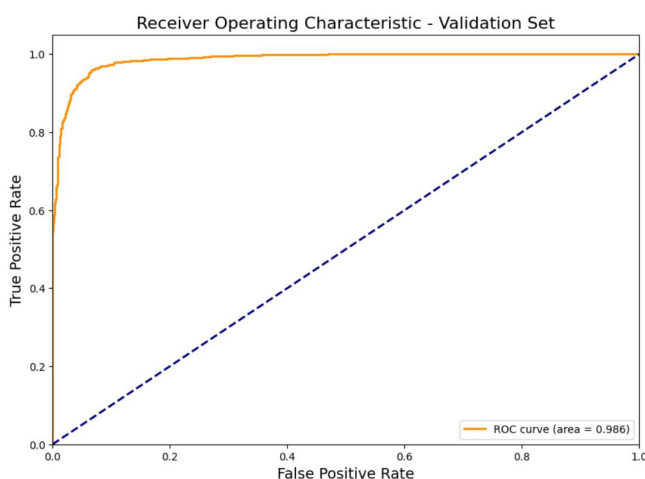


Fig. 5. ROC Curve for Binary Classification Model of Lesions

By analyzing the ROC curve and the AUC, we can determine the effectiveness of our model in distinguishing benign from malignant lesions, facilitating clinical decision-making and model refinement.

The analysis is crucial for assessing the overall effectiveness of the model and guiding future improvements. The experimental results demonstrate that the proposed two-stage approach outperforms baseline models and achieves superior classification accuracy on the ISIC2017 dataset. The integration of contrastive learning-based encoders with an EfficientNet classifier leverages the strengths of both methodologies, resulting in improved performance across all metrics. This approach not only enhances classification accuracy but also offers a robust framework for future research and practical applications in dermatology.

V. CONCLUSION

This study presents the effect of contrastive learning algorithm to enhance skin cancers classification. The experimental results confirm the effectiveness of the proposed two-stage approach in classifying skin diseases into benign and malignant categories. In conclusion, the proposed two-stage approach for skin disease classification represents a significant advancement in medical imaging, demonstrating the potential of deep learning technologies to enhance diagnostic processes. The method's ability to accurately classify skin lesions, combined with its potential applicability across various areas of medical imaging, underscores the importance of continued research and development in this field.

Expanding upon this foundation, future work should aim to refine these models further, enhance their interpretability, and validate their effectiveness in clinical settings to fully realize their potential in improving patient care.

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